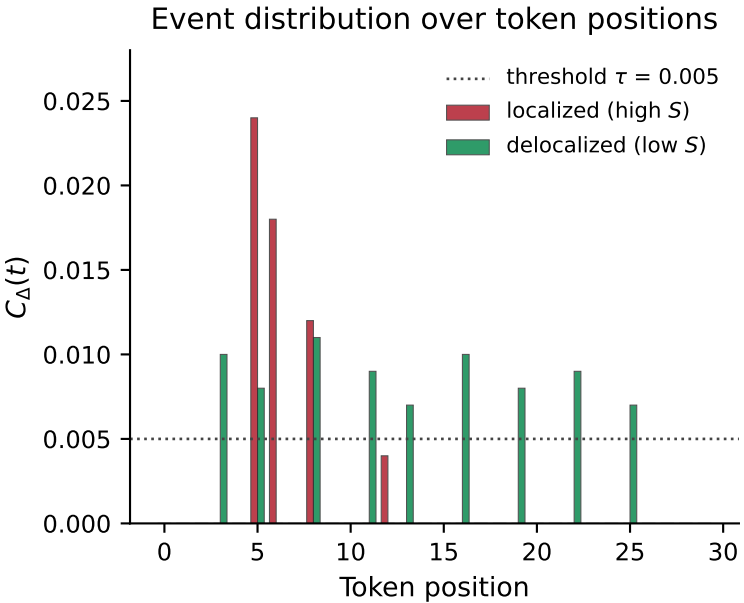
S as inverse participation ratio — physics grounding



$$S = M \cdot \mathrm{IPR}(\hat{e}) = \frac{\max_t C_\Delta(t)}{|\{t : C_\Delta(t) > \tau\}|}$$

(inverse participation ratio of the uniform event-location distribution)

Localized: $S = \frac{0.024}{4} = 0.0060$

Delocalized: $S = \frac{0.011}{9} = 0.0012$

ratio $\approx 5\times$ *— matches the cross-dataset* $d \approx 0.5$ *we observe*

Anderson 1958 · Edwards–Thouless 1972 · modern random-matrix localization